

WEEK 8

Week 8: Advanced Storage Concepts - Lab Guide

HPC & Cluster Computing

Objective: Configure a shared `/home` directory across the cluster. **Goal**: Allow a user to log into *any* node and see the same files.

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1. Prerequisites

- [ ] **Master Node**: Will act as the NFS Server.
- [ ] **Compute Nodes**: Will act as NFS Clients.
- [ ] **Networking**: Ensure firewall allows NFS traffic (Services: `nfs`, `mountd`, `rpc-bind`).

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2. Configure the NFS Server (Master)

We need to tell the Master to share its `/home` directory with the internal network.

2.1 Install Software

- [ ] Install the NFS utilities: `bash sudo dnf install nfs-utils -y`

Edit the exports file to define what to share and to whom.

- [ ] Open the configuration file:

```
bash sudo nano /etc/exports
```

- [ ] Add the following line (adjust IP range to match your cluster):

```
text /home 10.0.0.0/24(rw,sync,no_root_squash,no_all_squash)
```

2.3 Start Services and Firewall

Start the server and open the ports.

- [ ] Enable and start the NFS server:

```
bash sudo systemctl enable --now nfs-server
```

- [ ] Open the Firewall ports:

```
bash sudo firewall-cmd --add-service=nfs --permanent sudo firewall-cmd --add-service=rpc-bind --permanent sudo firewall-cmd --add-service=mountd --permanent sudo firewall-cmd --reload
```

2.4 Verify Export

- [ ] Check if the server is actively sharing: `bash exportfs -v # Output should show: /home 10.0.0.0/24(...)`

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3. Configure NFS Clients (Compute Nodes)



Now we need the compute nodes to mount the share. Since our nodes are **stateless** (Warewulf), we have two options:

1. **Runtime**: Run the mount command manually (good for testing).
2. **Persistent**: Add it to `/etc/fstab` inside the Warewulf Container or System Overlay (good for production).

For this lab, we will use the **Runtime** method using `pdsh` or manual entry to verify it works.

3.1 Install Software (On Compute Node)

Note: Your Warewulf container likely already has `nfs-utils`. If not, you must install it into the container image and rebuild.

3.2 Mount the Share

On `node01`:

- [ ] SSH into the node:

```
bash ssh node01
```

- [ ] Mount the share:

```
bash # Create a mount point (if /home isn't empty, mounting over it hides local contents) # Ideally /home is empty on the node image. sudo mount -t nfs 10.0.0.1:/home /home
```

- [ ] Verify it mounted:

```
bash df -h | grep home # Should see: 10.0.0.1:/home ...
```

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4. Verification Test (Interactive)

The ultimate test is file persistence. Perform this sequence to prove your setup works.

1. [ ] Create File on Master `bash echo "Hello from Master" > ~/test_file.txt`
2. [ ] Read File on Node01 `bash ssh node01 cat ~/test_file.txt # Expected Output: Hello from Master`
3. [ ] **3. Append on Node01** `bash ssh node01 "echo 'Hello from Node01' >> ~/test_file.txt"`
4. [ ] **4. Read File on Master** `bash cat ~/test_file.txt` **Did it work?** You should see both lines.

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5. Submission

Submit a screenshot of your terminal showing:

- [ ] `exportfs -v` on the Master.
- [ ] `df -h` on a Compute Node showing the NFS mount.